



NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering



SMART FARMER ASSISTANT

AI-Based Agriculture Support System



Submitted To:

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Section: 8D

Submission date: 22th June 2026

CSE4204 — Mobile Computing Lab

Week 04 Submission

System Design & Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

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1. Team Information

Field	Details
Team Name	CSE4204-8D-T03
Section	8D
Course	CSE4204 — Software Engineering Lab
Semester	Spring 2026
Institution	Leading University / University (Bangladesh)
GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant

1.1 Team Members

#	Name	Student ID	Role
1	Sakib Foysal Ejarder	11220320948	Team Leader / Backend
2	Md. Noyon Sheikh	11220320946	Frontend Developer
3	Sayed Tauhidul Islam	11220320950	AI / ML Engineer
4	Sayed Akib Osman	11220320974	Database / QA

2. Project Information

Project Overview

Smart Farmer Assistant is an AI-powered platform designed to assist farmers across Bangladesh in making informed agricultural decisions. Agriculture is a critical sector for Bangladesh' economy, yet farmers face persistent challenges including crop disease losses (20-30% annual yield loss), improper fertilizer usage, limited access to qualified agricultural extension officers, and lack of real-time market information.

This system integrates multiple AI and IoT technologies into a single responsive web application accessible from any smartphone or web browser, providing instant crop disease detection, 24/7 AI chatbot support, real-time weather forecasting with flood alerts, live market price tracking, and intelligent fertilizer recommendations.

Project Scope

- Current Features:
 - AI Crop Disease Detection via image analysis
 - AI-powered Chatbot (Gemini AI) with Bangla support
 - Real-time Weather Forecasting and Flood Alerts
 - Live Market Price Tracking by region
 - Intelligent Fertilizer Recommendation Engine
 - User Registration, Login, and Profile Management

- Disease Detection History and Chat History Tracking
- Admin Panel for system management

Target Users

User Type	Description
Farmers (Primary)	Small and medium-sized farmers who need crop disease diagnosis and farming guidance
Agricultural Workers	Farm laborers who can benefit from disease detection and weather alerts
Admin Users	System administrators responsible for managing users and updating market data

3. UI/UX Design - Wireframes and Screens

The following section describes the major user interface screens of the Smart Farmer Assistant. The design follows a mobile-first responsive approach, ensuring optimal experience across all devices.

Design Philosophy

Simplicity: Clean, intuitive interface requiring minimal learning curve for rural farmers

Accessibility: Large touch targets, readable fonts, and bilingual (English/Bangla) support

Responsiveness: Works seamlessly on mobile phones, tablets, and desktop browsers

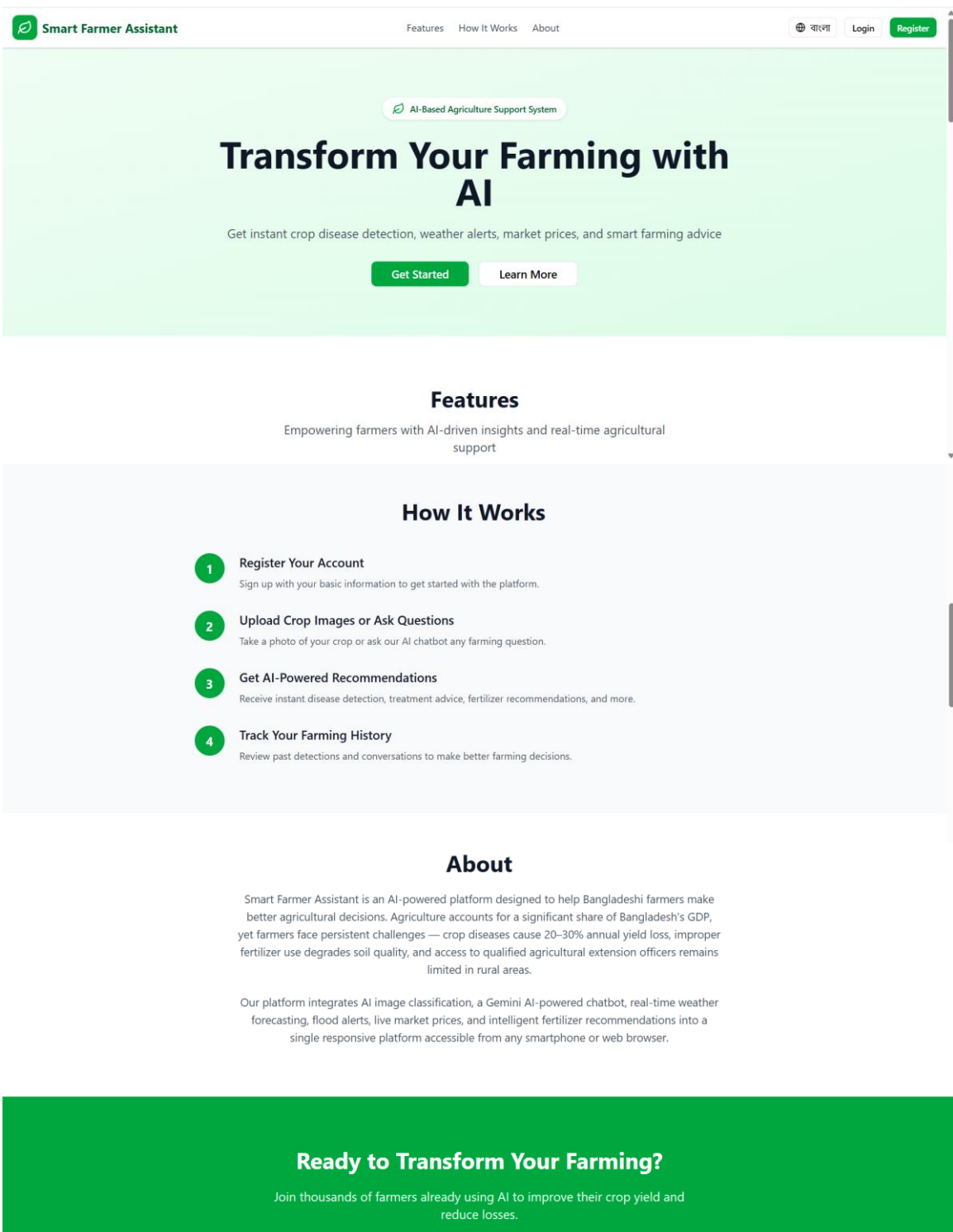
Performance: Fast loading times and minimal data consumption for low-bandwidth areas

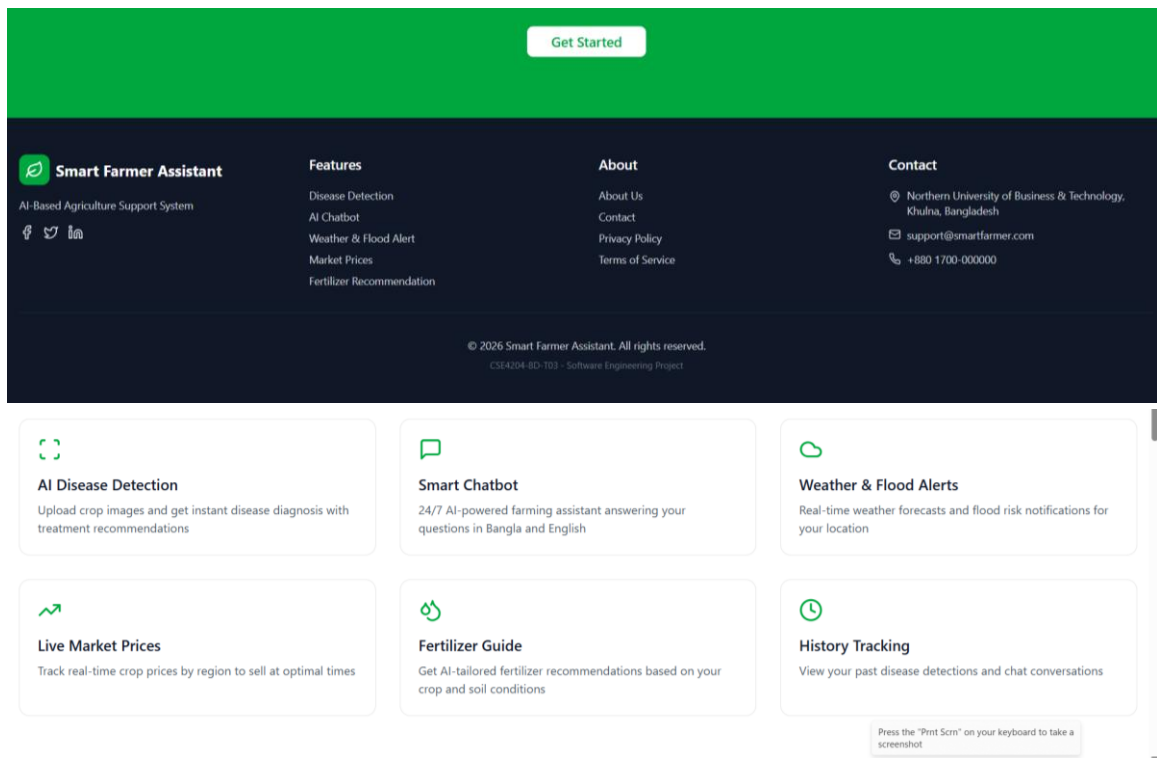
Trust: Professional design that builds confidence in AI recommendations

3.1. Landing/Home Page

Welcome page showcasing key features of the system. Contains navigation menu, hero section with call-to-action buttons (Get Started, Learn More), and feature highlights.

- Key Elements:
 - Navigation Bar
 - Hero Section
 - Features Overview
 - Call-to-Action Buttons
 - Footer

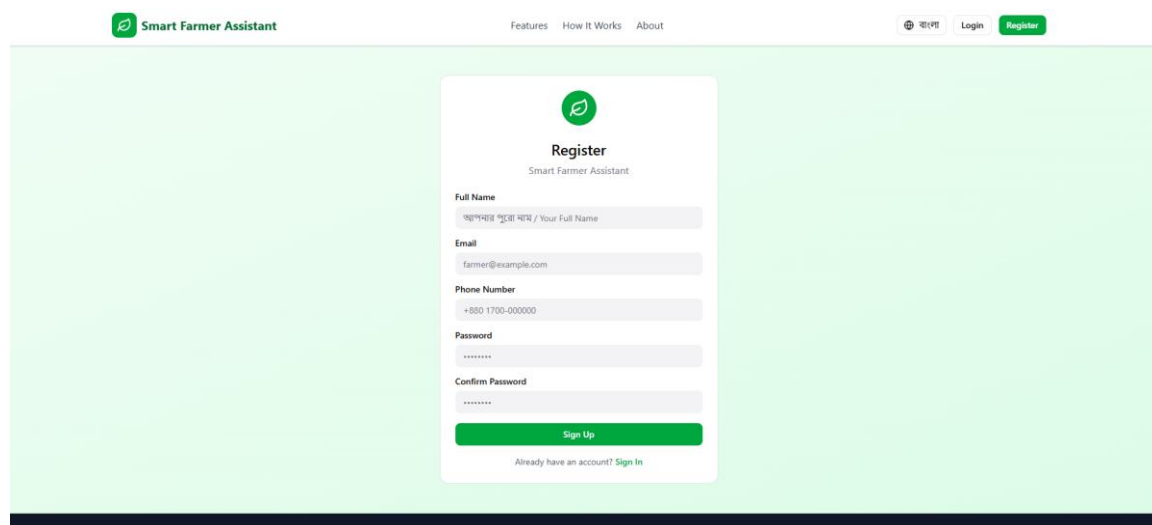




3.2. Registration Page

User registration form for new farmers. Collects full name, email, phone number, and password. Includes validation and error handling. Provides link to login page for existing users.

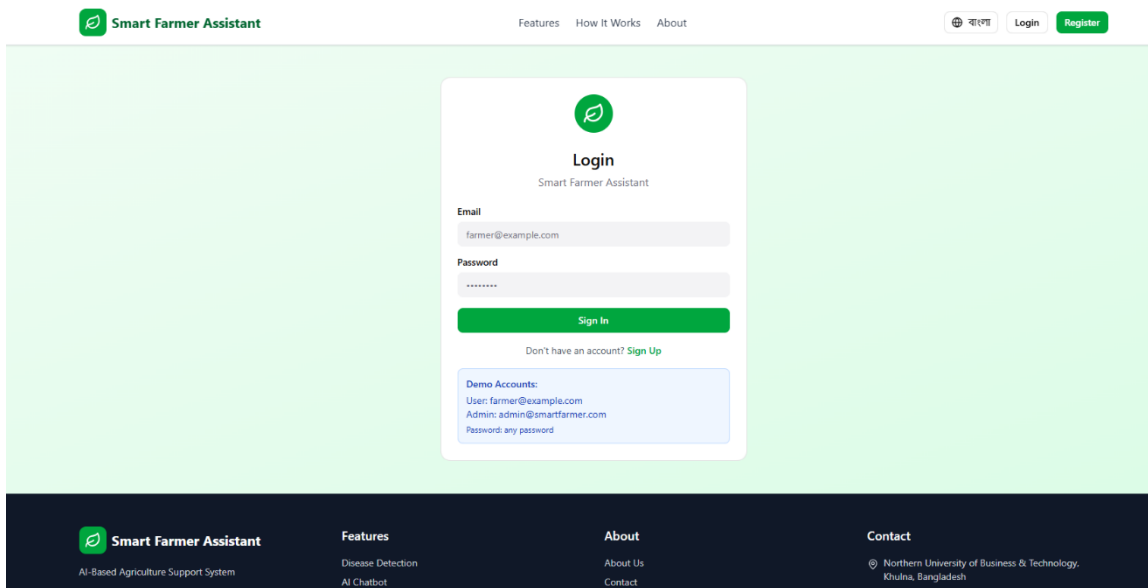
- Key Elements:
 - Full Name Field
 - Email Field
 - Phone Number Field
 - Password Fields (with confirmation)
 - Sign Up Button
 - Login Link



3.3. Login Page

Authentication page with email and password fields. Includes demo account credentials for testing. Shows both Bangla and English text.

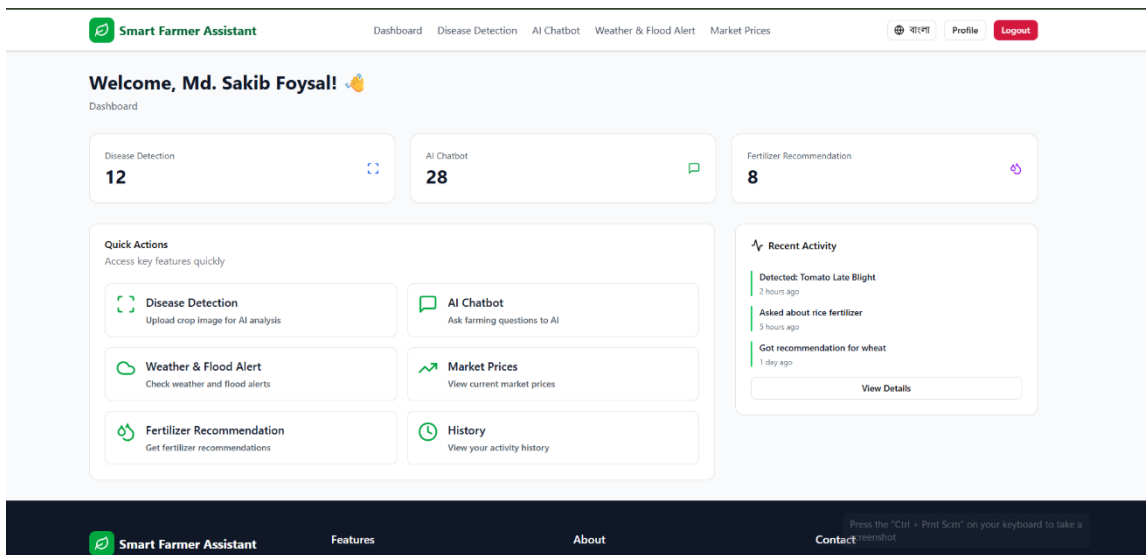
- Key Elements:
 - Email Input
 - Password Input
 - Sign In Button
 - Registration Link
 - Demo Credentials Section



3.4. Dashboard

Main landing page after login. Shows welcome message with user's name, quick statistics (disease detections, chatbot messages, fertilizer recommendations), quick action buttons, and recent activity feed.

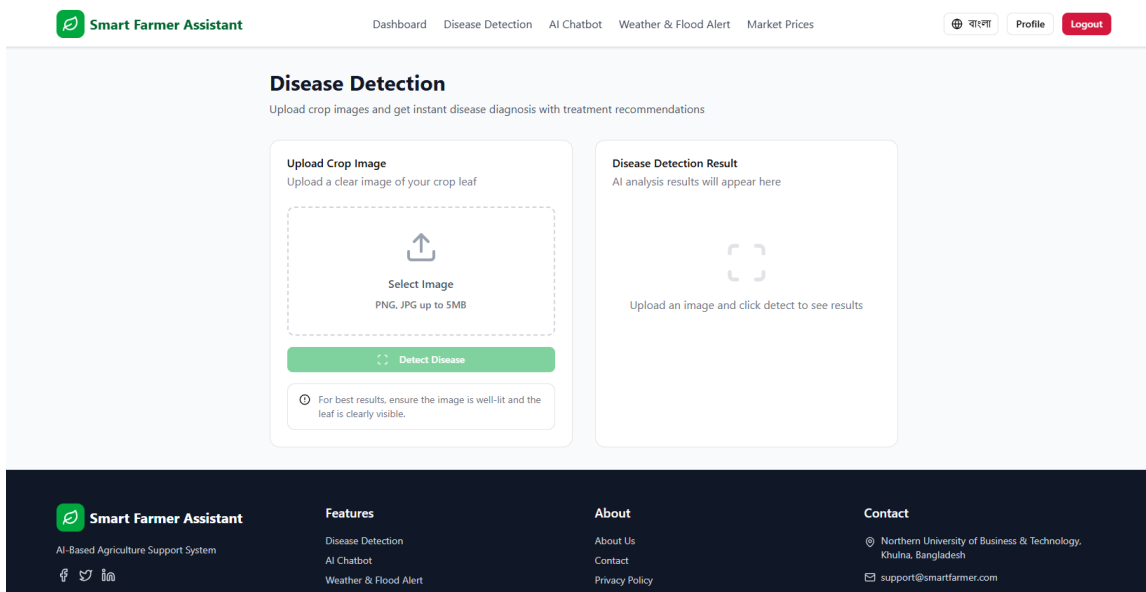
- Key Elements:
 - Welcome Message
 - Stats Cards
 - Quick Actions
 - Recent Activity
 - Navigation Menu



3.5. Disease Detection Page

Two-panel layout: left side for image upload with drag-and-drop support, right side for displaying AI analysis results. Shows disease name, confidence score, treatment recommendations.

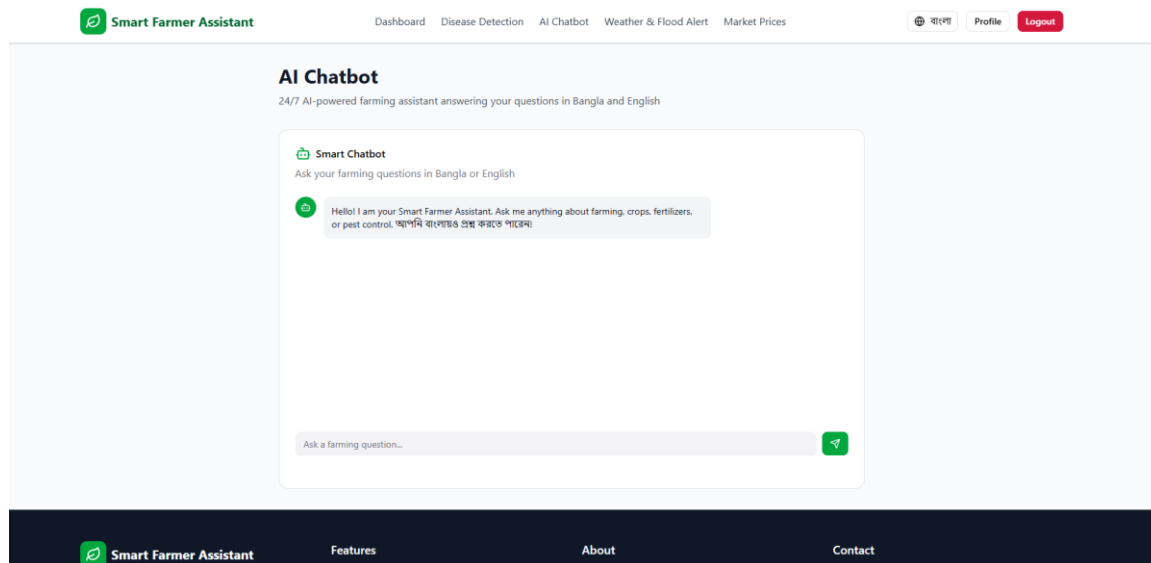
- Key Elements:
 - Image Upload Area
 - Detect Button
 - Results Panel
 - Disease Name
 - Confidence Score
 - Treatment Advice



3.6. AI Chatbot Interface

Chat-based interface for asking farming questions. Supports both Bangla and English. Shows conversation history. Displays AI responses with contextual farming advice.

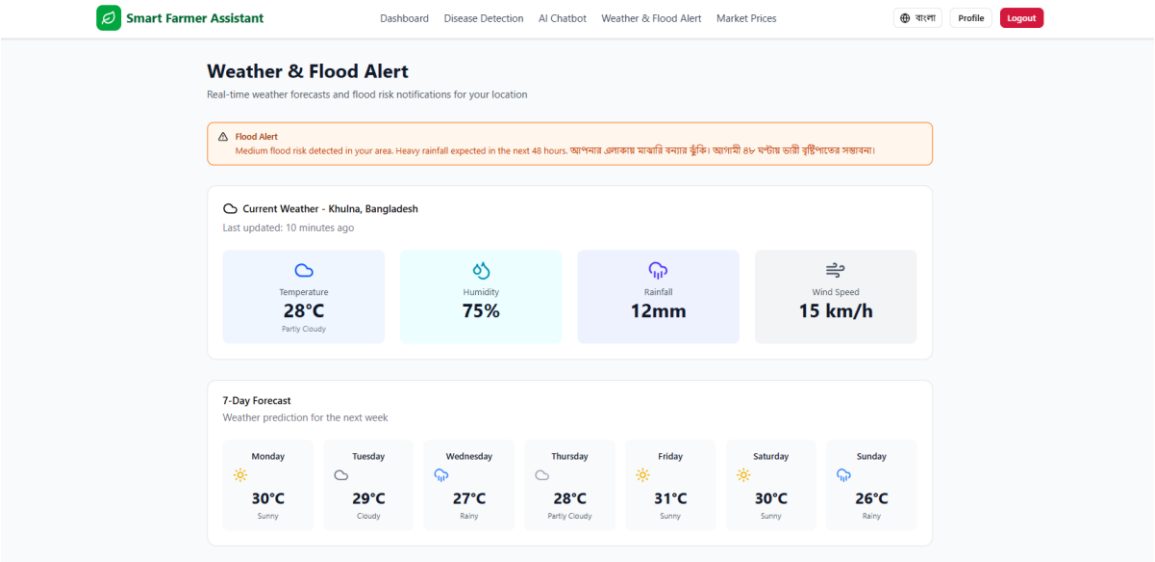
- Key Elements:
 - Chat History
 - Question Input
 - Send Button
 - Language Toggle
 - AI Response Display



3.7. Weather & Flood Alert Page

Displays current weather with temperature, humidity, rainfall, wind speed. Shows 7-day forecast with daily weather icons. Includes flood risk alerts at the top.

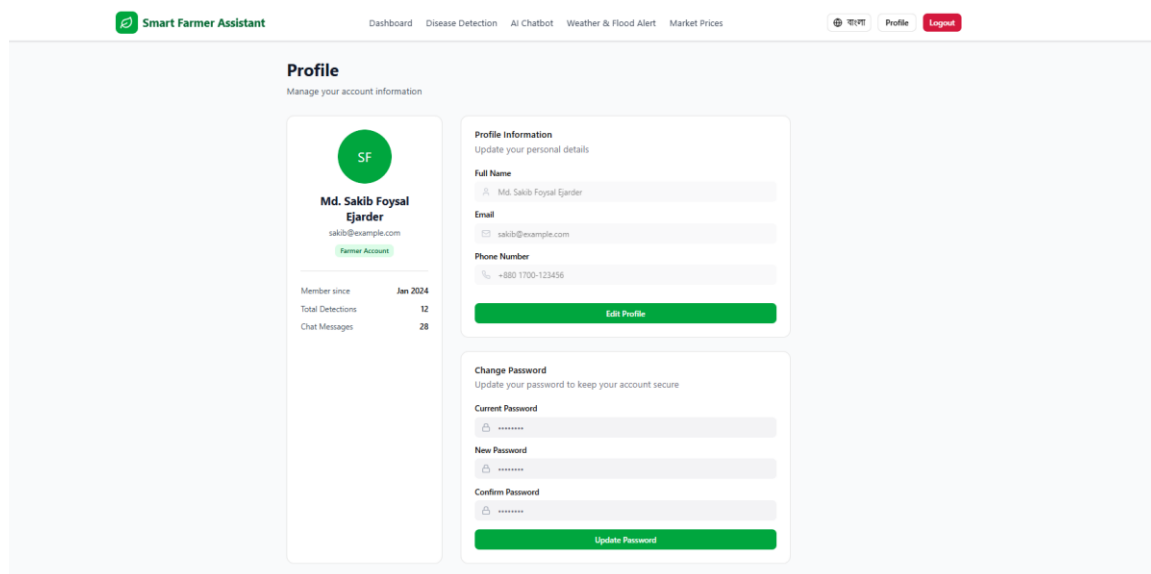
- Key Elements:
 - Flood Alert Box
 - Current Weather Cards
 - 7-Day Forecast
 - Weather Icons
 - Location Display



3.9. User Profile Page

User account management page. Shows profile avatar with initials, personal information, account statistics (member since, total detections, chat messages), edit profile button, and change password section.

- Key Elements:
 - Profile Avatar
 - User Name
 - Email
 - Phone
 - Account Stats
 - Edit Profile Button
 - Change Password Section



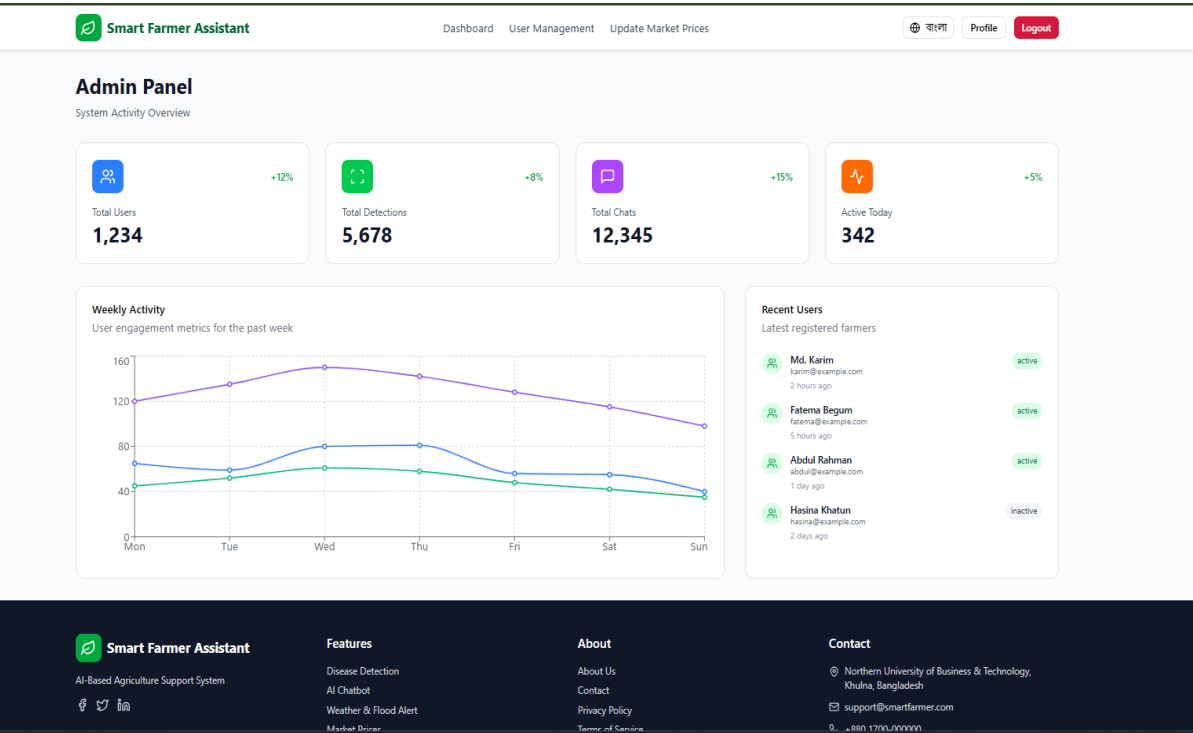
The screenshot displays the 'Smart Farmer Assistant' user profile page. The header includes the logo and navigation links: Dashboard, Disease Detection, AI Chatbot, Weather & Flood Alert, and Market Prices. User navigation options are shown as 'বাংলা', 'Profile', and 'Logout'.

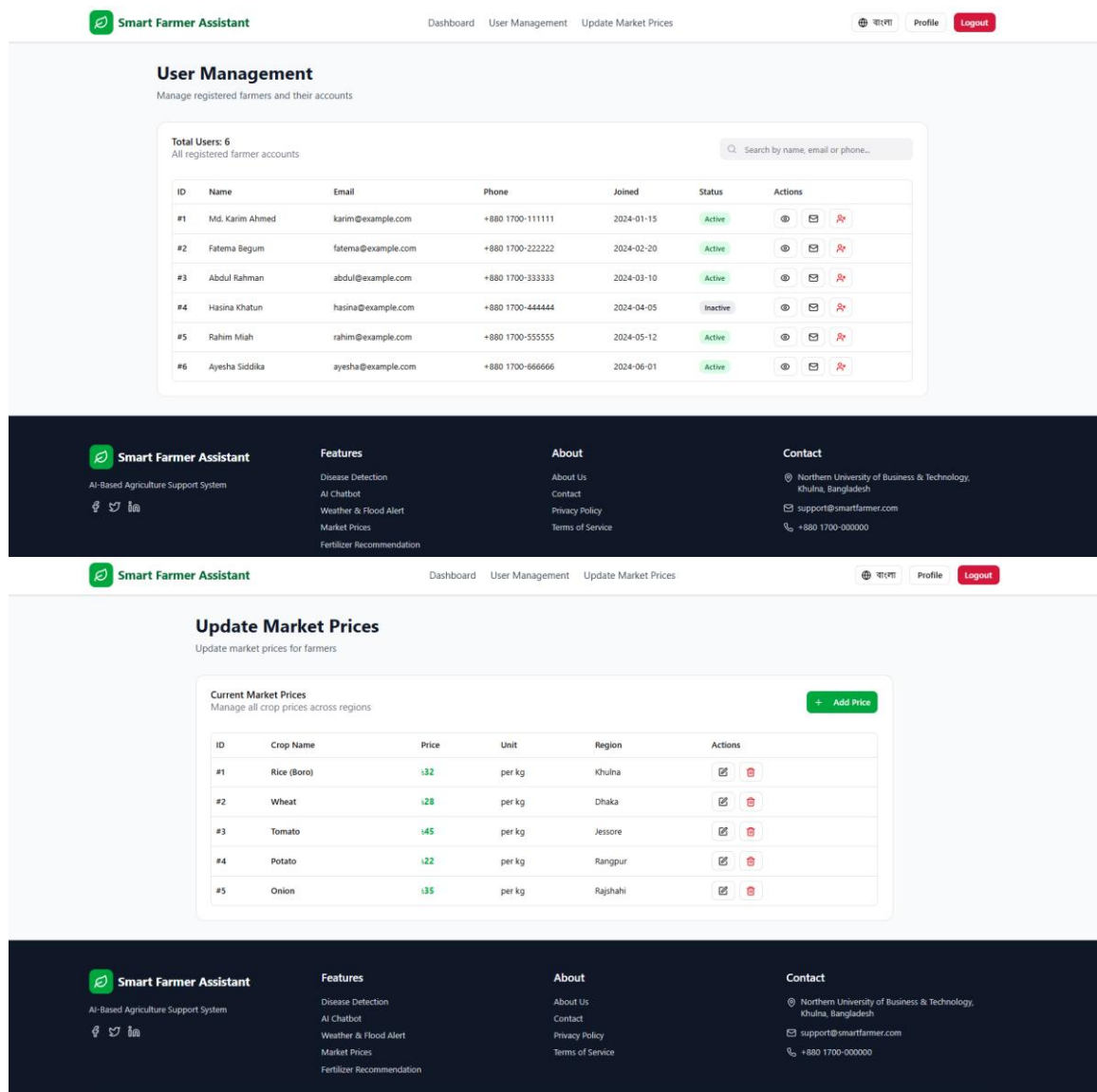
The main content area is titled 'Profile' with the subtitle 'Manage your account information'. It is divided into three sections:

- Profile Information:** Includes a circular avatar with initials 'SF', the name 'Md. Sakib Foysal Ejarder', email 'sakib@example.com', and a 'Farmer Account' tag. Account statistics are listed: Member since Jan 2024, Total Detections 12, and Chat Messages 28.
- Profile Information Form:** Contains input fields for Full Name (Md. Sakib Foysal Ejarder), Email (sakib@example.com), and Phone Number (+880 1700-123456), with an 'Edit Profile' button.
- Change Password Form:** Includes fields for Current Password, New Password, and Confirm Password, with an 'Update Password' button.

3.10. Admin Page

Main control panel for administrators after login. Displays a welcome message with the admin's name, system overview statistics (total users, disease reports, fertilizer recommendations, chatbot interactions), quick management actions, notifications, and recent system activities for efficient monitoring and administration.





Design Features

- Bilingual Support (Bangla & English) for accessibility
- Green color scheme (#1DA758) for agricultural/environmental theme
- Responsive grid layout optimized for mobile (320px) to desktop (1920px)
- Consistent navigation bar with language toggle and user menu
- Large, touch-friendly buttons for rural users
- Clear form validation and error messages
- Loading indicators for AI processing
- Confirmation dialogs for destructive actions